Identities, Multiple Angles, and Exact Solutions

Simplify with an identity, convert a multiple angle, and list every exact solution on the interval.

These problems mix three jobs that look alike on the page but need different first moves. Decide which one you are being asked for before you write anything:

- **Simplify / verify.** No equals sign to solve — rewrite everything in sines and cosines, or reach for the Pythagorean identity, and reduce.
- **Multiple angle.** A $2x$ (or 2θ) appears. Replace it with a double-angle form *chosen* so that only the function you already know survives.
- **Exact solution set.** There is an interval. Reduce to one trigonometric function, factor, then read *every* unit-circle angle in the interval — exact values, not decimals.

Show the identity you used at each step. Leave all answers exact.

1. Simplify to a single trigonometric function:

$$\frac{1-\cos^2 x}{\sin x}$$

Answer: ________________

2. Simplify to a single trigonometric function:

$$\sec x - \sec x \sin^2 x$$

Answer: __________________

3. Rewrite $\cos 2x$ so that it contains $\sin x$ and no other trigonometric function.

Answer: __________________

4. Find every exact solution of

$$2\sin x - 1 = 0$$

on the interval $[0, 2\pi)$.

Answer: ____________________

5. Suppose $\sin\theta = \dfrac{3}{5}$ and θ lies in quadrant I, so that $\cos\theta = \dfrac{4}{5}$.
Find the exact value of $\sin 2\theta$.

Answer: ____________________

6. Simplify to a single term:

$$\tan x \cos x + \sin x$$

Answer: ________________

7. Find every exact solution of

$$2\cos^2 x - \cos x - 1 = 0$$

on the interval $[0, 2\pi)$. *Hint:* it factors as a quadratic in $\cos x$.

Answer: ________________

8. Simplify to a single trigonometric function:

$$\frac{\sin 2x}{1+\cos 2x}$$

Answer: ____________________

9. Combine and simplify. Your answer should contain no fractions inside fractions:

$$\frac{1}{1-\sin x}+\frac{1}{1+\sin x}$$

Answer: ____________________

10. Find every exact solution of

$$\sin 2x = \sin x$$

on the interval $[0, 2\pi)$. *Hint:* expand the double angle first, then bring everything to one side and factor — do not divide by $\sin x$.

Answer: ________________

11. Suppose $\cos\theta = -\dfrac{5}{13}$ and θ lies in quadrant II.
Find the exact value of $\cos 2\theta$.

Answer: ________________

12. **Challenge.** Find every exact solution of

$$2\cos^2 x + \sin x = 1$$

on the interval $[0, 2\pi)$.

This one needs all three moves: an identity to reach a single function, a factorisation, and a complete sweep of the interval.

Answer: ____________________